Test Project Desktop
HovRail Kiosk

Introduction

HovRail Kiosk is a C# desktop application designed exclusively for passengers to manage railway bookings through a simple, self-service interface. The app allows users to explore the railway network and make pre-reservations, creating a streamlined booking experience.

With two primary features—route visualization and self pre-reservation—the app offers a comprehensive view of available routes, displaying each network of connections. Passengers can then make self pre-reservations for their chosen routes, securing seats without an immediate payment. Payment is handled separately through a later process, making the app ideal for users with prepaid options or accounts set up for post-booking payments.

Contents

This Test Project consists of the following files:

1. TP_ITSSB_Desktop_SELEKNAS_2024.pdf
2. HovRailKiosk.sql

Deliverables

You are expected to submit your Desktop Project as a compressed (zip) file. You do not need to include your database file, as the assessment will use the original provided database. Failure to meet these requirements may affect your assessment results.

Entity Relationship Diagrams

The following are Entity Relationship Diagram (ERD) of the HovRail Kiosk.

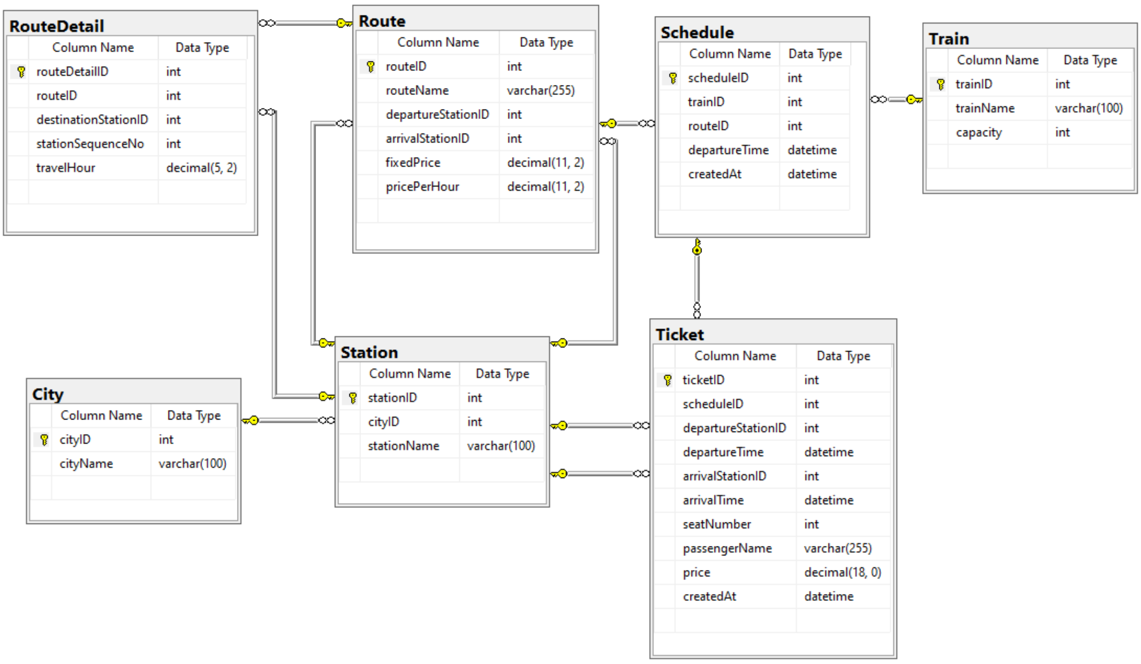

Figure 1. Entity Relationship Diagram of HovRail Kiosk

Application Requirements

1. Main Form

- Figure 2 shows the main form displayed when the application starts, with the date and time updating every second
- There are two main menus:
  a. View Rail Network
  b. Book Ticket
- Clicking the button redirects the user to the corresponding form.

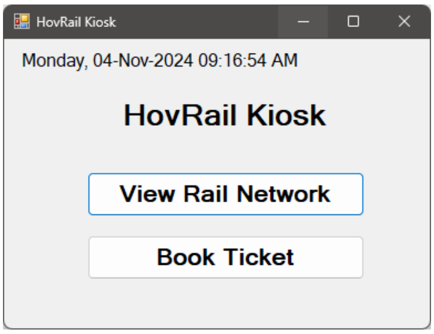

Figure 2. Main Form of HovRail Kiosk

2. View Rail Network Form

- Figure 3 shows the View Rail Network Form. This form can be used by user to observe all routes available.

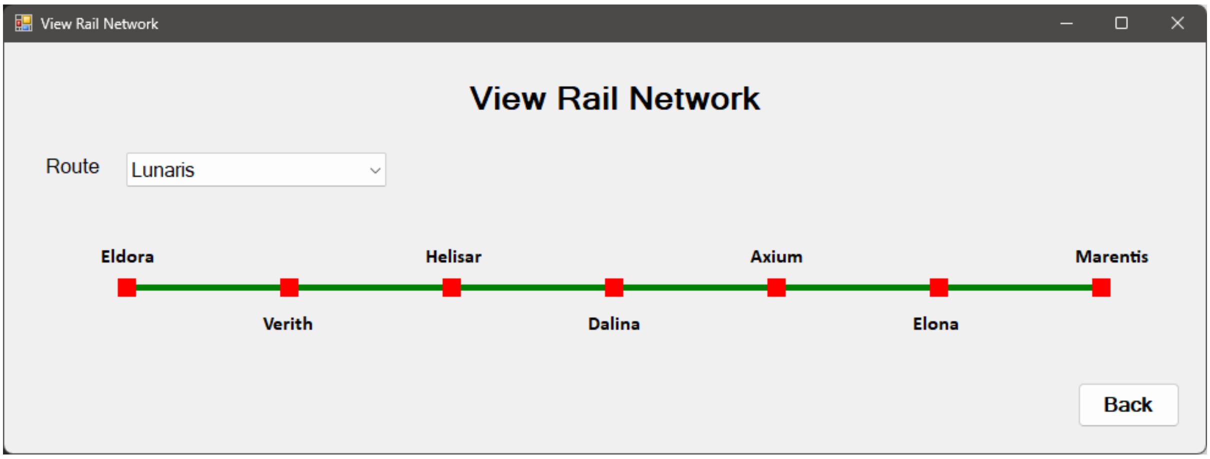

Figure 3. View Rail Network Form

- The route dropdown lists all available routes. Changing the selected option will update the route visualization accordingly.
- The route will be visualized as a single green line, with each station stop marked by a small red rectangle. Station labels will be positioned, starting at the top of the line, moving downward, and repeating this pattern until the end. Ensure that each station mark is evenly spaced along the line. Please refer to the visual above for clarity.
- User can click on the back button to navigate to the Main Form.

3. Book Ticket Form

- Each booking is limited to one passenger and applies only to a single route, with no transfer options available between the selected departure and arrival stations.
- The Book Ticket Form is a single form divided into three distinct process sequences. Users cannot move freely between these sequences; they can only navigate back and forth using the previous and next buttons.
- Users can cancel the process at any time by clicking the cancel button, which will redirect them to the Main Form. Ensure that the user is warned about the implications of this cancellation.

- The first sequence (Figure 4) involves selecting a schedule by choosing the departure station, arrival station, and date. The departure and arrival station must be different.
- Bookings must be made at least one day in advance and up to three days before departure. The data grid view will show all available schedule based on user requirements. Here are important points to note:
  - ✓ **Price Calculation:** The price will be calculated using the following formula:

$$\boldsymbol{Price} = \boldsymbol{Fixed\ Price} + (\boldsymbol{Travel\ hour\ Departure\ to\ Arrival} * \boldsymbol{Price\ Per\ Hour})$$

  - ✓ **Available Seats:** The available seats represent the number of vacant seats for the desired requirement. If there are no vacant seats, the corresponding row will be highlighted in red and cannot be selected.
  - ✓ **Schedule Selection:** Users must first select a schedule to proceed to the next step of the booking process.

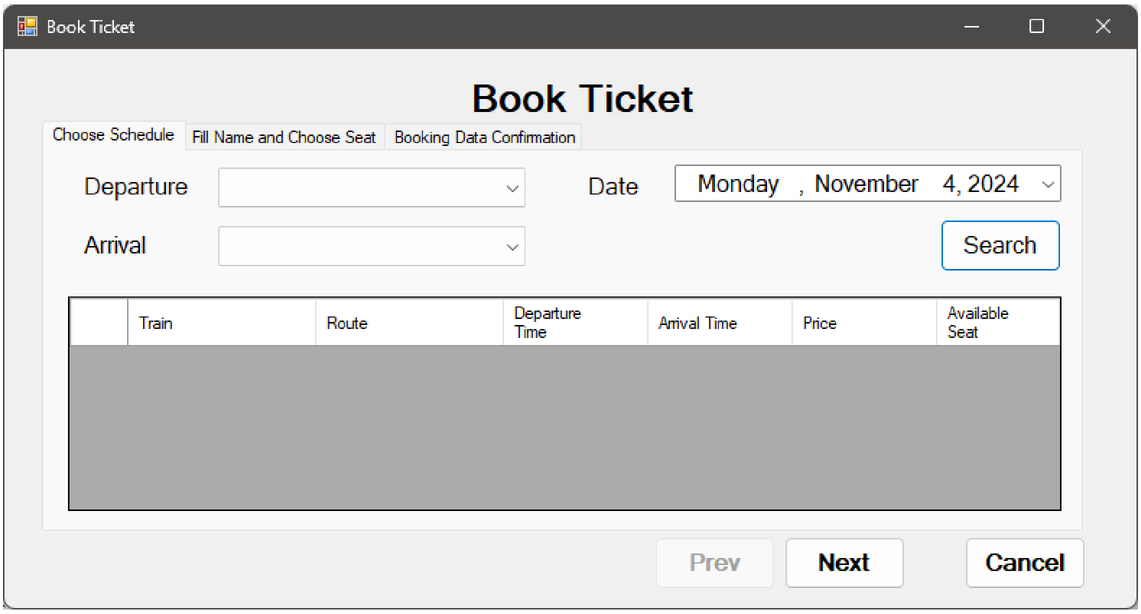

Figure 4. Book Ticket Form (Choose Schedule)

- The second sequence (Figure 5) requires the user to fill in the passenger's name, which is mandatory.
- Next, the user needs to select a seat to book. The seat display is based on the train's total capacity, with booked seats highlighted in red and unavailable for selection. Initially, a random seat will be assigned by the system, but the user can choose any available seat to change their selection afterward. The selected seat will be highlighted as yellow.

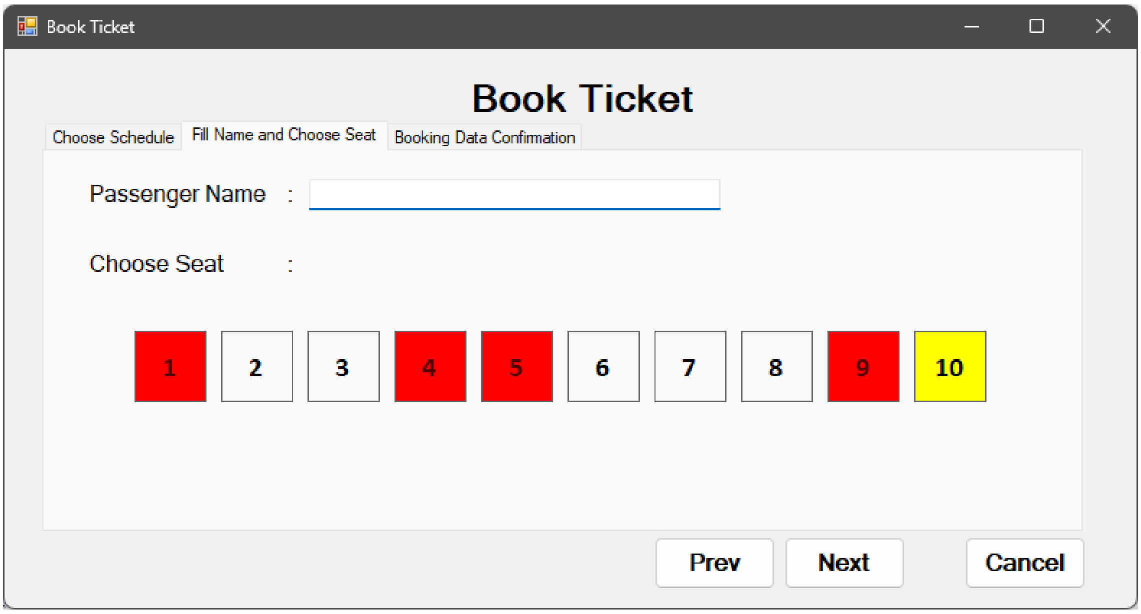

Figure 5. Book Ticket Form (Fill Name and Choose Seat)

- The third sequence (Figure 6) requires the user to confirm their booking details. All previously filled information will be displayed here. Clicking the submit button will save the booking data and redirect the user to the Main Form.

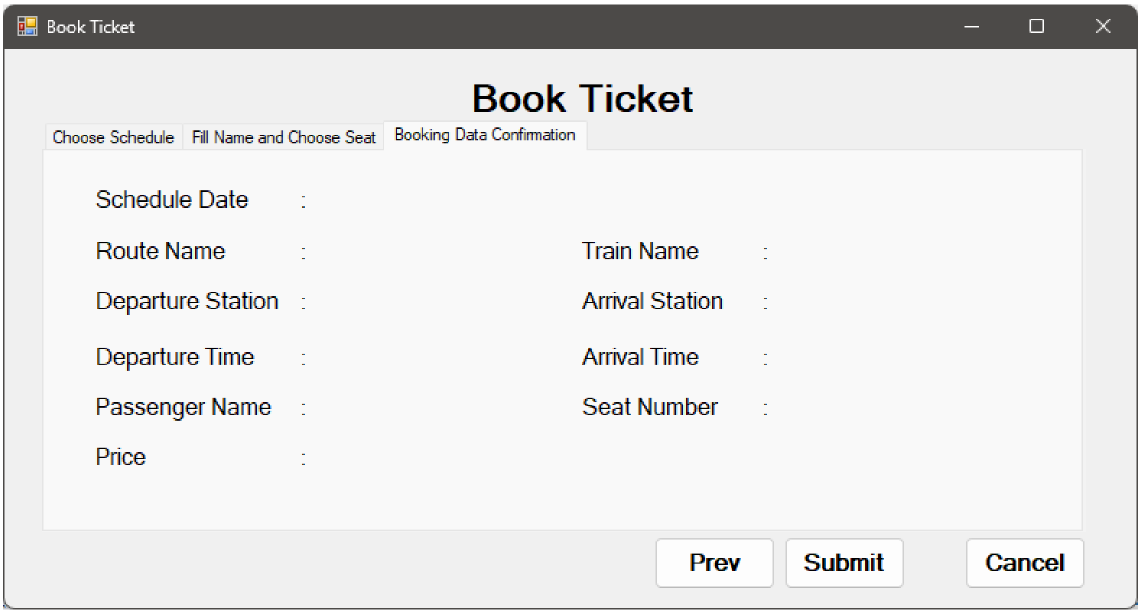

Figure 6. Book Ticket Form (Booking Data Confirmation)